CODECHECK certificate 2025-023

https://codecheck.org.uk/register/certs/2025-023/

Table 1: CODECHECK summary

Item	Value
Title	*A calibrated optogenetic toolbox of stable zebrafish opsin lines*
Author(s)	Paride Antinucci, Adna Dumitrescu, Charlotte Deleuze, Holly J Morley, Kristie Leung, Tom Hagley, Fumi Kubo, Herwig Baier, Isaac H Bianco, Claire Wyart
Reference	https://doi.org/10.7554/eLife.54937
Repository	https://github.com/codecheckers/certificate-2025-023
Codechecker(s)	Linus Dexter Hackel
Date of check	2026-01-15
Summary	Figures 4, 5, 8 and 9 could be reproduced partially. More precisely, the figures, which were produced by the scripts inlcude a range of graphs and there is always one graph which matches a graph in the related figure in the paper. However some produced figures also look completely different to those in the paper like e.g. `figures/figure4_1.pdf`, whose graphs aren't included in figure 4 of the paper enitrely. Furthermore, like already said, each script produces a range of graphs, but only one seems to appear in the related figure of the paper. I am not sure why that is and also how that graph is selected. There was no documentation how Figures 1, 2, 3, 6 and 10 where produced or with which scripts, so they couldn't be reproduced.

Table 2: Summary of output files generated

File	Comment	Size (b)
`Gapfree_AP_stim.csv`	The .csv file containing the gap free AP stimulation.	1870
`figure4_1.pdf`	Manuscript Figure 4 (Trace: 2019_03_16_0000, Opsin: Cheriff)	16548
`figure4_2.pdf`	Manuscript Figure 4 (Trace: 18n270027_1, Opsin: Chrimson)	82263
`figure4_3.pdf`	Manuscript Figure 4 (Trace: 2019_03_19_0038, Opsin: CoChR)	63693
`figure5_1.pdf`	Manuscript Figure 5 (Trace: 2019_03_19_0055, Opsin: GtACR1)	78091
`figure5_1.pdf`	Manuscript Figure 5 (Trace: 18d130007_5, Opsin: Chrimson)	78091

File	Comment	Size (b)
`figure8_1.pdf`	Manuscript Figure 8 (Trace: 183060053_1, Opsin: NpHR)	235116
`figure8_2.pdf`	Manuscript Figure 8 (Trace: 2019_08_01_0064, Opsin: GtACR1)	213717
`figure9_1.pdf`	Manuscript Figure 9 (Trace: 186280022_1, Opsin: NpHR)	245474
`figure9_2.pdf`	Manuscript Figure 9 (Trace: 2019_02_25_0035, Opsin: GtACR1)	248817
`figure9_3.pdf`	Manuscript Figure 9 (Trace: 2019_08_01_0067, Opsin: GtACR1)	179181
`figure9_4.pdf`	Manuscript Figure 9 (Trace: 189040010_1, Opsin: NpHR)	94949
`figure9_5.pdf`	Manuscript Figure 9 (Trace: 2019_01_25_0007, Opsin: GtACR1)	121586

Summary

Figures 4, 5, 8 and 9 could be reproduced partially. More precisely, the figures, which were produced by the scripts inlcude a range of graphs and there is always one graph which matches a graph in the related figure in the paper. However some produced figures also look completely different to those in the paper like e.g. `figures/figure4_1.pdf`, whose graphs aren't included in figure 4 of the paper enitrely. Furthermore, like already said, each script produces a range of graphs, but only one seems to appear in the related figure of the paper. I am not sure why that is and also how that graph is selected. There was no documentation how Figures 1, 2, 3, 6 and 10 where produced or with which scripts, so they couldn't be reproduced.

CODECHECKER notes

Environment

Setting up the environment took a bit of time, as older versions of Python and older Dependencies needed to be properly installed, but it is all very well documented in the `README` file what dependencies need to be installed and with which version.

Since I didn't want a new window for the plot results of MatPlotLib, I descided to put the following sequence at the end of each script, so the created figure is automatically saved to the `outputs/figures/` directory.

```
figure_number = 9

figure_id = int(input('Please enter the Figure ID.\n\n'))

plt.savefig(f"figures/figure{figure_number}_{figure_id}.pdf")
```

Script Errors

In the script `Excitatory_Opsin_Current_Clamp.py` the filepath for the trace contained a typo in line 452. It was: `'Analysis_output/Single_trace_data/CC_excitatory/2019_03_19_0055.csv'` but it should have been `'Analysis_output/Single_Trace_data/CC_excitatory/2019_03_19_0055.csv'`. So instead of `Trace` the url inlcuded `trace`. This is just a small error, but it still took me some minutes in figuring out, where the `FileNotFoundError` could be coming from, as the file appeared to be there.

Similar to the first error, in the script `Inhibitory_Opsin_Current_Clamp.py` the filepath for the CC inhibitory opsin master sheet wasn't correct and needed to be changed in the lines 400, 417, 423. Here the problem was, that it was the full file path to the authors private directory, so it needed to be changed to my directory. My recommendation is, to only use relative file paths, such that these errors can be avoided. Different to the first error though, the file didn't exist entirely, so it had to be newly created with the same `csv-Header` as the file `'Analysis_output/VC_inhibitory_opsin_master.csv'`. After these two fixes, the script worked perfectly.

In the script `Inhibitory_Opsin_CC_Long_AP_Inhibit.py` the line 424 had to be changed from an array of the length 7 to an array of the length 2, as this ws what was given in the documentation and also what the code afterwards expected. The code was therefore changed from:

```
LED_max_V_user = [input('pulse_1:  \n'), input('pulse2:  \n'), input('pulse3:  \n'), input('pulse4:  \n'),
input('pulse5:  \n'), input('pulse6:  \n'), input('pulse7:  \n')]
```

to

```
LED_max_V_user = [input('pulse_1:  \n'), input('pulse2:  \n')]
```

Recommendations to the authors

TODO

Citing this document

Linus Dexter Hackel (2026). CODECHECK Certificate 2025-023. Zenodo. https://codecheck.org.uk/register/certs/2025-023/

About CODECHECK

This certificate confirms that the codechecker could independently reproduce the results of a computational analysis given the data and code from a third party. A CODECHECK does not check whether the original computation analysis is correct. However, as all materials required for the reproduction are freely availableby following the links in this document, the reader can then study for themselves the code and data.

About this document

This document was created using codecheck-py (a Python-base template for creating CODECHECK certificates). The CODECHECK details are filled into a jupyter notebook which is then converted into Markdown via nbconvert. Afterwards it gets converted into Typst using cmarker and then into PDF using Typst.

License

Manifest files

CSV files

`Analysis_output/Gapfree_AP_stim.csv`

Author comment: *The .csv file containing the gap free AP stimulation.*

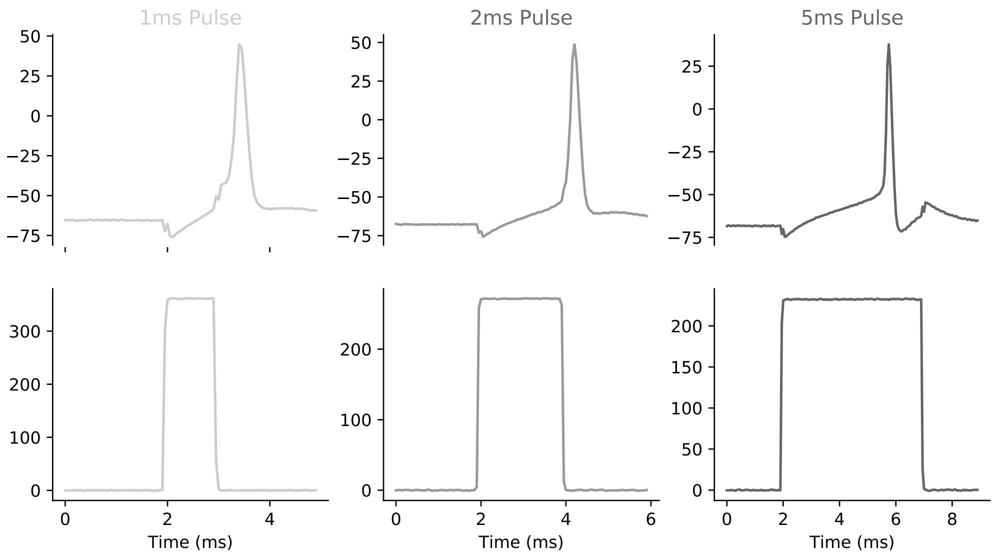
Column summary statistics:

	count	mean	std	min	25%	50%	75%	max
0	8	3.5000	2.4495	0.0000	1.7500	3.5000	5.2500	7.0000

Figures

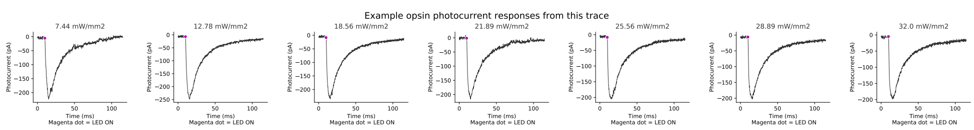
`figures/figure4_1.pdf`

Author comment: *Manuscript Figure 4 (Trace: 2019_03_16_0000, Opsin: Cheriff)*



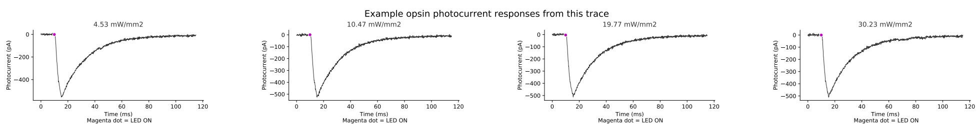
`figures/figure4_2.pdf`

Author comment: *Manuscript Figure 4 (Trace: 18n270027_1, Opsin: Chrimson)*



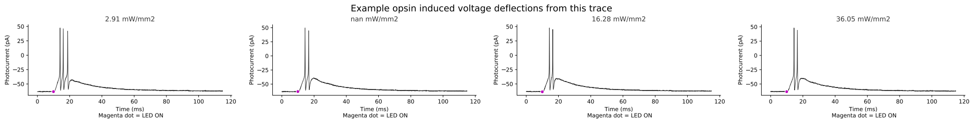
`figures/figure4_3.pdf`

Author comment: *Manuscript Figure 4 (Trace: 2019_03_19_0038, Opsin: CoChR)*



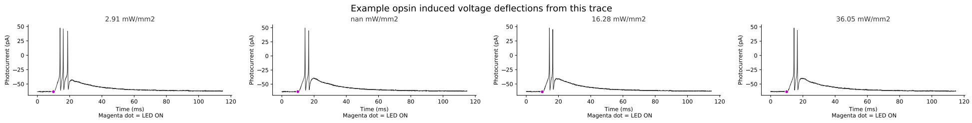
`figures/figure5_1.pdf`

Author comment: *Manuscript Figure 5 (Trace: 2019_03_19_0055, Opsin: GtACR1)*



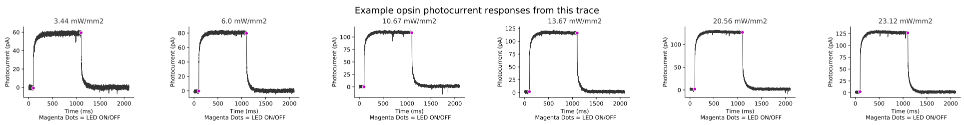
figures/figure5_1.pdf

Author comment: *Manuscript Figure 5 (Trace: 18d130007_5, Opsin: Chrimson)*



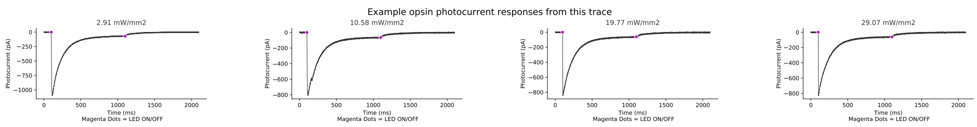
figures/figure8_1.pdf

Author comment: *Manuscript Figure 8 (Trace: 183060053_1, Opsin: NpHR)*



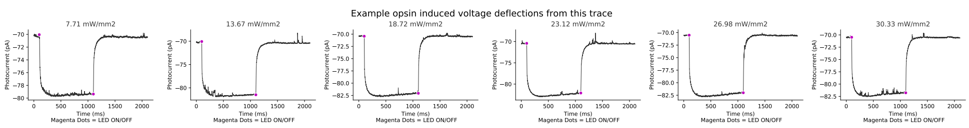
figures/figure8_2.pdf

Author comment: *Manuscript Figure 8 (Trace: 2019_08_01_0064, Opsin: GtACR1)*



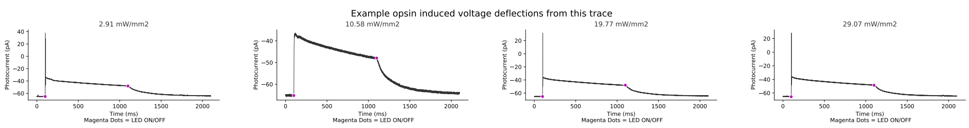
figures/figure9_1.pdf

Author comment: *Manuscript Figure 9 (Trace: 186280022_1, Opsin: NpHR)*



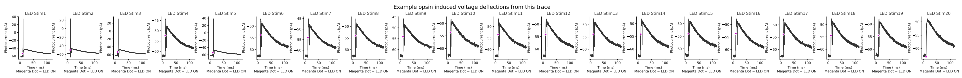
figures/figure9_2.pdf

Author comment: *Manuscript Figure 9 (Trace: 2019_02_25_0035, Opsin: GtACR1)*



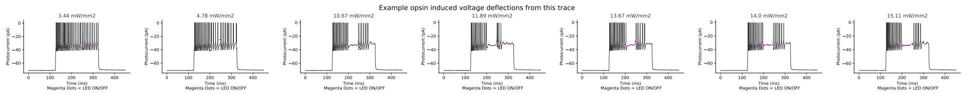
figures/figure9_3.pdf

Author comment: *Manuscript Figure 9 (Trace: 2019_08_01_0067, Opsin: GtACR1)*



figures/figure9_4.pdf

Author comment: *Manuscript Figure 9 (Trace: 189040010_1, Opsin: NpHR)*



figures/figure9_5.pdf

Author comment: *Manuscript Figure 9 (Trace: 2019_01_25_0007, Opsin: GtACR1)*